Topic 16 — Oceans

Sources baked in: NCERT Class 11 (Oceans); **PWOnlyIAS Udaan** (ocean-floor relief); **Drishti** To-the-Point (ocean currents); **ForumIAS UPPSC** oceanography map; Lucent match-lists; UNCLOS; **Ghatnachakra** World Geography Hydrosphere–Seas–Currents–Salinity–Tides–Trenches–Straits–Canals–Coral (CB–115+ / CB–156+); UPPCS Prelims 2018–2025 **Weight:** ★★★ — warm vs cold currents, trench–ocean match, salinity belt, Suez, UNCLOS miles, Maitri–Bharati; teach full ocean map **Last verified:** September 2026 — seas/straits coast desk + Hormuz CA pass **Current Affairs:** **Strait of Hormuz 2026** crisis (oil chokepoint); BBNJ 2023; Arctic observer status

Current Affairs (this topic)

Year	Fact	Why it matters	Source
Feb 2026–ongoing	After US–Israel strikes on Iran (from **28 Feb 2026**), Iran moved to control / restrict the **Strait of Hormuz**; tanker traffic collapsed; mines, boardings and attacks on merchant ships (नावाध्यक्ष) were reported	World's top oil–LNG chokepoint in the news	Reuters / IEA / CRS
2025 baseline	About **20 mb/d** of oil products (~**25%** of seaborne oil) and ~**20%** of world **LNG** (mainly Qatar + UAE) normally transit Hormuz	Scale of any closure	IEA
Static geography	Hormuz joins **Persian (फ़ारसी) Gulf** ↔ **Gulf of Oman**; shores are **Iran (north)** and **Oman (Musandam, south)**; UAE sits on the approaches	Country / strait MCQs	Atlas
Bypass note	Only **Saudi Arabia** and the **UAE** have meaningful pipeline routes that can skip Hormuz; Iraq, Kuwait, Qatar, Bahrain and Iran lean almost entirely on the strait	Who is trapped if closed	IEA
Demand side	Most Hormuz oil goes to **Asia** (China, India, Japan, Korea); India is a major importer through this gate	India linkage	EIA / visualisers
Other 2026 themes	Dual blockade talk (Iran on the strait / US on Iranian ports), insurance spikes, and later escort / corridor politics kept oil prices elevated	Chokepoint = price risk	Reuters / Brookings
Static	**BBNJ 2023** high-seas biodiversity treaty — India has signed; UNCLOS **12 / 24 / 200 nm** unchanged	Law of the sea	UN
Static	India Arctic Policy **2022**; Arctic Council **observer** since **2013**; Deep Ocean Mission / Samudrayaan / Blue Economy (नीली अर्थव्यवस्था)	Polar + deep ocean	MEA / MoES

N.0 Framework — The World Ocean

Oceans cover most of Earth and set climate, fish wealth and geopolitics.

- The **hypsographic profile** runs land → **shelf** → **slope** → **rise or trench** → **abyssal plain**, with **mid-ocean ridges** building new crust.
- **Horizontal** surface motions are wind-driven **currents** and **gyres**. **Vertical** density motion is the **thermohaline** conveyor.
- **Temperature, salinity and density** must be read as one linked set — density drives deep flow.
- **Waves** (constructive / destructive) are separate from **tides**. Do not merge them.
- **UNCLOS** zones (12 / 24 / 200 nm) are legal, not physical depth words like neritic / hadal.

Order of this chapter

1. Oceans, seas and floor morphology.
2. Trenches, currents, temperature and salinity.
3. Tides, deposits, straits and canals.
4. Law of the sea, fisheries and polar stations.

Teach the **full ocean map** — next year can ask any standard NCERT current, trench or deposit, not only Agulhas or Mariana.

N.1 Oceans & Seas

Five named oceans plus many seas and gulfs. Area order and Indian Ocean identity are high-yield.

Ocean	Area & Depth	Salinity / Features	Key Historical & Geographic Facts
Pacific	~166.24 million sq. km; Avg depth ~**4,200 m** (deepest ocean)	~35‰	Named *Pacific* (meaning calm or peaceful) by **Ferdinand Magellan** during world circumnavigation. Mariana Trench is deepest point (11,033 m). Contains >20,000 islands, mostly volcanic or coral.
Atlantic	~86.56 million sq. km; Avg depth ~3,646 m	High salinity in subtropics (~37‰)	Derived from **Atlas** (Titan in Greek mythology). Formed when Gondwanaland rifted (South America and Africa separated) and is still widening. Has the **longest coastline** and is the **busiest ocean for global commerce/trade** (connects industrialised W. Europe & NE North America).
Indian	~73.43 million sq. km; Avg depth ~**3,741 m** (deeper on average than the Atlantic)	Variable (~35–37‰ in Arabian Sea, lower in Bay of Bengal)	**Only ocean named after a country** (India). Greeks called it Erythraean Sea. Current circulation completely reverses seasonally with Monsoon winds. Contains major continental islands (Madagascar, Sri Lanka) and volcanic/coral islands (Mauritius, Seychelles, Maldives, Lakshadweep, Andaman & Nicobar).

Ocean	Area & Depth	Salinity / Features	Key Historical & Geographic Facts
Southern	~20.3 million sq. km	Low	Encircles Antarctica; carries the continuous Antarctic Circumpolar Current (West Wind Drift).
Arctic	~13.23 million sq. km; Avg depth ~**1,500 m** (shallowest ocean)	**Least salinity (~20‰)**	Smallest ocean, centered on the North Pole within the Arctic Circle. Largely ice-covered year-round with widest continental shelves.

Seas - Named seas include the Arabian Sea, Bay of Bengal, Andaman (अंडमान) Sea, Red Sea, Mediterranean, Black Sea, North Sea, Baltic Sea and Caribbean Sea. - East Asian and Pacific marginal seas include the South China, East China, Yellow, Japan, **Okhotsk**, Bering, Coral, Tasman and Java seas. - Polar and island seas include the Ross, Weddell, Arafura, Celebes, Banda and Philippine seas. - The **Sargasso Sea** sits in the North Atlantic gyre and has **no land boundary**. - The **Caspian** and **Aral** are **lakes**, not true seas.

Seas desk — learn by “what is it between?”

Do not memorise only coordinates. Read each map as **who sits on which side**, then learn the connector strait or current.

South China Sea



- The **South China Sea** is a **western Pacific** marginal sea in Southeast Asia.
- Picture the frame: **China** on top, **Vietnam** on the left, the **Philippines** on the right, and **Borneo / Malaysia** below.
- Core coastal set: China, Taiwan, Vietnam, Philippines, Malaysia, Brunei, and parts of Indonesia.
- It is often the **deepest** option among common seas (~**5377 m**).
- Island traps inside the basin include the **Paracel** and **Spratly** groups.

Bering Sea



- The **Bering Sea** is a **North Pacific** marginal sea between **Russia** and **Alaska (USA)**.
- Easy cue: **Bering = between Russia and Alaska**.
- The **Bering Strait** links the Bering Sea to the **Arctic Ocean**.

Mediterranean Sea



- The **Mediterranean** sits between **Europe** (north) and **Africa** (south), with **Asia** on the eastern side.

- Atlantic Ocean → **Strait of Gibraltar** → Mediterranean.
- It is the classic large enclosed sea in MCQs.
- **Syria does border the Mediterranean**, but **Syria does not border the Black Sea**.
- **Jordan does not** open to the Mediterranean; Jordan's only sea tip is the **Gulf of Aqaba** (Red Sea).

Black Sea



- Ring the coasts: **north Ukraine**, **east Russia + Georgia**, **south Türkiye**, **west Bulgaria + Romania**.
- Strait chain (high-yield order): **Black Sea** → **Bosporus** → **Sea of Marmara** → **Dardanelles** → **Aegean** → **Mediterranean**.
- **Syria** is a Mediterranean country only — never put it on the Black Sea list.

Sea of Azov



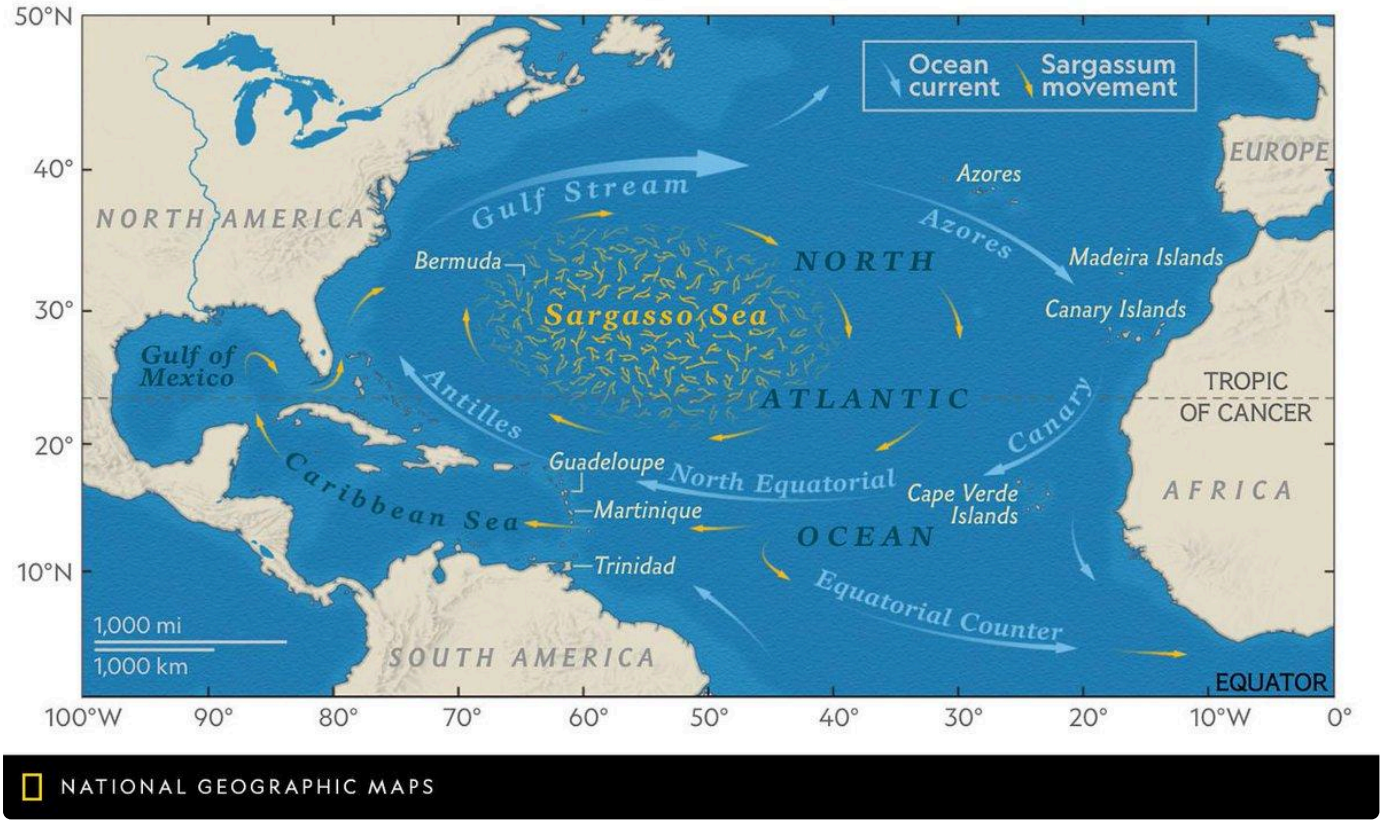
- The **Sea of Azov** sits **north of the Black Sea**, mainly between **Ukraine** and **Russia**.
- It opens to the Black Sea through the **Kerch Strait**.
- Cue: **Azov = small sea parked above the Black Sea**.

Tyrrhenian Sea



- The **Tyrrhenian Sea** is **west of Italy**.
- Frame: **Italy** to the east, **Corsica** and **Sardinia** to the west, **Sicily** to the south.
- Cue: **T = Tyrrhenian = to the west of Italy**.

Sargasso Sea



- The **Sargasso Sea** lies in the **North Atlantic**, roughly around **Bermuda**.
- It has **no coastline**; currents form the boundary (NOAA: only sea without a land shore).
- Current frame: **west Gulf Stream**, **north North Atlantic Current**, **east Canary Current**, **south North Atlantic Equatorial Current**.
- Named for **Sargassum** weed floating in the gyre.

Aral Sea



- The **Aral Sea** sits in **Central Asia** between **Kazakhstan** and **Uzbekistan**.
- Despite the name “sea”, it is a **lake**.
- Feeders: **Syr Darya** from the northeast and **Amu Darya** from the south.
- Heavy irrigation diversion has shrunk the lake into separate remaining water bodies.

Red Sea



- The **Red Sea** separates **Africa** (Egypt, Sudan, Eritrea, Djibouti) from the **Arabian Peninsula** (Saudi Arabia, Yemen).
- North: **Suez Canal** links the Red Sea to the **Mediterranean**.
- South: **Bab-el-Mandeb** opens to the **Gulf of Aden** and then the **Indian Ocean**.
- Form: **axial trough / rift** between the African and Arabian plates — long, narrow, and deep for its width.

Quick “between what?” table

Sea	Just remember this
South China Sea	China – Vietnam – Philippines – Borneo
Bering Sea	**Russia** ↔ **Alaska**
Mediterranean	**Europe** ↔ **Africa**, Asia east

Sea	Just remember this
Black Sea	Ukraine – Russia – Georgia – Türkiye – Bulgaria – Romania
Sea of Azov	**North of Black Sea**, via Kerch Strait
Tyrrhenian	**West of Italy**, Italy–Sardinia–Corsica–Sicily
Sargasso	**North Atlantic; no coastline; currents define it**
Aral	**Kazakhstan + Uzbekistan; shrinking lake**
Red Sea	**Africa ↔ Arabia; rift**

Five map chains to revise first

1. **Black Sea + Azov + Bosporus + Dardanelles**
2. **Mediterranean + Tyrrhenian + Adriatic + Aegean + Ionian**
3. **Red Sea + Gulf of Aden + Bab-el-Mandeb + Suez**
4. **South China Sea + surrounding countries**
5. **Aral Sea + Syr Darya + Amu Darya**

Türkiye's four (चातुर्याम) waters — clock method

Picture Türkiye on the map and read the coasts like a clock face.

Direction	Sea	Easy cue
North (12)	**Black Sea**	Cold wall **above** Türkiye
South (6)	**Mediterranean**	Warm "middle sea" **below** Türkiye
West (9)	**Aegean Sea**	Greek-islands side
North-west pocket	**Sea of Marmara**	Tiny inland sea between Black and Aegean

One north→south chain through the straits: Black Sea → **Bosphorus** → Sea of Marmara → **Dardanelles** → Aegean → Mediterranean.

- "North of Turkey" always means the **Black Sea**, never the Caspian.
- "Turkey is located between…" usually means the big north–south frame: **Black Sea and Mediterranean**.
- The **Aegean** and **Marmara** are western connectors. They are not Caspian neighbours.

West → East seas ladder

Say aloud: **"My Black Cat Ate."**

Letter	Sea
M	**Mediterranean**
B	**Black**

Letter	Sea
C	**Caspian**
A	**Aral**

That is the west-to-east order across Eurasia. The **Caspian** and **Aral** are **lakes**, not true seas, but the ladder still appears in MCQs.

Gaza Strip

The **Gaza Strip** is a narrow coastal belt on the **Mediterranean**. It does not face the Red Sea. The **Dead Sea** is inland in the Jordan Rift, so it is not Gaza's open coast.

Baltic coast ring

Walk clockwise from Denmark around the Baltic:

Denmark → Germany → Poland → Lithuania → Latvia → Estonia → Russia → Finland → Sweden.

Say: **"Danish Germans planted three Baltics; Russia, Finland and Sweden close the ring."**

- **Norway** faces the North Sea / Norwegian Sea / Atlantic.
- **Norway does not** sit on the Baltic coast.

Jordan and the Mediterranean

In South-West Asia, **Syria**, **Lebanon** and **Israel** open to the Mediterranean. **Jordan does not.**

Jordan's only sea outlet is the **Gulf of Aqaba** (a Red Sea arm), shared with Israel, Egypt and Saudi Arabia.

Important seas — who touches the coast?

Use this desk when a stem asks "which country borders / does not border" a sea.

Sea / water body	Coastal countries (core list)	Classic "does NOT" trap
Mediterranean	Spain, France, Monaco, Italy, Slovenia, Croatia, Bosnia & Herzegovina, Montenegro, Albania, Greece, **Türkiye**, **Syria**, **Lebanon**, **Israel**, Egypt, Libya, Tunisia, Algeria, Morocco (+ island states)	**Jordan**, Chad, Iraq, Saudi Arabia
Black Sea	Russia, Ukraine, Romania, Bulgaria, **Türkiye**, Georgia	**Syria** (Med only)
Caspian (lake)	Russia, Kazakhstan, Turkmenistan, Iran, Azerbaijan	**Türkiye**, Uzbekistan
Aral (lake)	Kazakhstan, Uzbekistan	Azerbaijan / Russia as pair
Red Sea	Egypt, Sudan, Eritrea, Djibouti, Saudi Arabia, Yemen, Israel, **Jordan** (Aqaba tip)	Jordan–Mediterranean
Persian / Arab Gulf	Iran, Iraq, Kuwait, Saudi Arabia, Bahrain, Qatar, **UAE**	**Oman** (faces Gulf of Oman / Arabian Sea)

Sea / water body	Coastal countries (core list)	Classic “does NOT” trap
Gulf of Oman	Oman, Iran, UAE (approaches)	Calling it “inside” the Persian Gulf
Baltic Sea	Denmark, Germany, Poland, Lithuania, Latvia, Estonia, Russia, Finland, Sweden	**Norway**
North Sea	UK, Norway, Denmark, Germany, Netherlands, Belgium, France	Mixing with Baltic list
Arabian Sea	India, Pakistan, Iran, Oman, Yemen, Somalia, Maldives approaches	Treating Persian Gulf as identical
Bay of Bengal	India, Bangladesh, Myanmar, Sri Lanka approaches	—
South China Sea	China, Taiwan, Vietnam, Philippines, Malaysia, Brunei, Indonesia (parts)	—

Aegean / Marmara reminder: the Aegean is shared mainly by **Greece** and **Türkiye**. The Sea of Marmara is almost entirely a **Turkish** inland sea.

Inline PYQ — UPPCS Prelims 2025, Q78

With reference to the Indian Ocean, which of the following statements is/are correct?

1. It was called by the Ancient Greeks as the Erythraean Sea.
2. It is spread on either side of the Equator.

A. Only 2

B. Neither 1 nor 2

C. Both 1 and 2

D. Only 1

Show answer

Ans: C

Gulfs / bays

A gulf or bay is an arm of the sea into the land. ask: which sea does it open into, and which strait guards the mouth?

Name	Where	Opens into	Note
Persian / Arab Gulf	Between Iran and Arabia	Gulf of Oman → Arabian Sea	Mouth at the **Strait of Hormuz** (oil chokepoint)
Gulf of Oman	SE of Hormuz	Arabian Sea	Outer side of Hormuz — **not** the same as the Persian Gulf

Name	Where	Opens into	Note
Gulf of Aden	Between Yemen and Somalia	Arabian Sea	Inner side of **Bab-el-Mandeb**, leading into the Red Sea
Gulf of Suez	NW arm of **Red Sea** (Egypt)	Red Sea	**Not** the Suez Canal lakes. The canal starts from this gulf toward the Mediterranean
Gulf of Aqaba	NE arm of Red Sea	Red Sea	Shared by Israel, Jordan, Egypt and Saudi Arabia, parallel to the Gulf of Suez
Gulf of Mexico	S of USA, E of Mexico	Atlantic (via Florida / Yucatan straits)	Caribbean neighbour with the Loop Current and major oil fields
Gulf of Guinea	West Africa (Nigeria–Ghana (घन) coast)	Atlantic	Warm **Guinea Current** and the Niger delta
Gulf of California (Sea of Cortez)	Between Baja and mainland Mexico	Pacific	Pacific arm, not Atlantic
Gulf of Alaska	S of Alaska	Pacific	Warm **Alaska Current**
Gulf of Thailand	Cambodia / Thailand / Vietnam	South China Sea	Shallow gulf, not part of the Indian Ocean
Gulf of Tonkin	N Vietnam / S China	South China Sea	Northern South China Sea arm
Gulf of Khambhat / Cambay	Gujarat, between Kathiawar and mainland	Arabian Sea	India's classic **tidal-energy** coast
Gulf of Kutch	N of Kathiawar (Gujarat)	Arabian Sea	Second tidal site, with reefs
Gulf of Mannar	Between TN and Sri Lanka	Indian Ocean	Reefs and biosphere reserve (जैवमंडल आरक्षित क्षेत्र). **Palk Strait** is the narrower passage to the north
Hudson Bay	NE Canada	Atlantic via Hudson Strait	Huge inland sea, not ranked as a true ocean
Bay of Fundy	Nova Scotia–New Brunswick, Canada	Atlantic	**Highest tidal range in the world** (not Khambhat)
Gulf of Carpentaria	N Australia	Arafura Sea	Shallow tropical gulf
Bay of Biscay	W France / N Spain	Atlantic	Rough seas on Europe's continental shelf
Gulf of Bothnia	Between Sweden and Finland	Baltic	Low-salinity Baltic arm
Gulf of Finland	Finland / Estonia / Russia (St Petersburg)	Baltic	Eastern Baltic arm

Name	Where	Opens into	Note
Bay of Bengal	E of India	Indian Ocean	Named a **bay** but treated as a sea; cyclone belt

Suez trap - The **Gulf of Suez** is the north-west **natural** arm of the **Red Sea**. - The **Suez Canal** is the dug channel from the Mediterranean to that gulf. - Canal lakes (Manzala → Timsah → Great Bitter → Little Bitter) are **inside the canal**, not the same as the Gulf of Suez. - Full canal teaching is in **N.8** below.

N.1A Hydrosphere budget · datum · OMT

The **hydrosphere** is Earth's water in oceans, ice, groundwater, lakes, rivers and vapour. Oceans cover about **71%** of Earth's surface — roughly **two-thirds** (ते-भागा) of the planet. The world ocean is one connected body, named as five basins: Pacific > Atlantic > Indian > Southern > Arctic.

Fresh water (IAS framing)

- Salt water ≈ **96.5%** of all water. Fresh ≈ **2.5%**.
- Usable fresh for humans ≈ **less than 1%** of total Earth water.
- Of fresh water: ice/glaciers ≈ **68.7%**, groundwater ≈ **30.1%**, surface (lakes/rivers) ≈ **0.3%**, other ≈ **0.9%**.
- So after ice, the largest remaining fresh store is **groundwater** (not lakes/rivers). Rivers and lakes together hold **less** fresh water than groundwater.

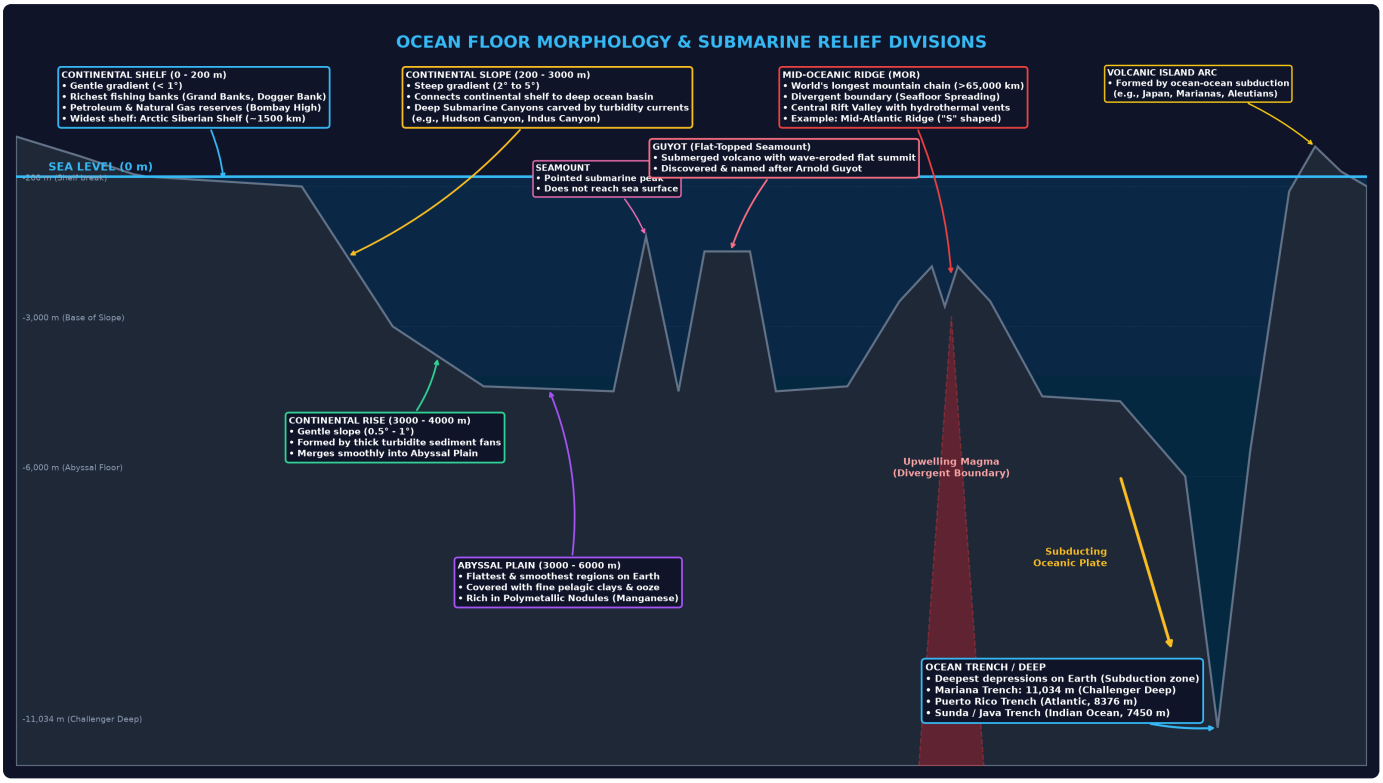
Most water is held in the salty ocean. Of the small fresh share, **ice sheets and glaciers** dominate. Groundwater is the next big store. Liquid surface water in rivers and lakes is tiny by comparison, which is why irrigation and cities lean heavily on aquifers.

Datum line: horizontal reference at **mean sea level** from which heights and depths are measured — not the International Date Line and not the prime meridian.

Ocean Mean Temperature (OMT) is measured to the **26°C isotherm** depth (often ~50–100 m). The SW Indian Ocean January–March mean is about **59 m** in the IAS key, not 129 m. OMT from January–March helps judge whether Indian monsoon rainfall will sit above or below the long-term mean — a better pair than SST alone.

Gravitational water in soil sits in macropores and drains under gravity. Tension is typically **below ~1/3 atmosphere**.

N.2 Ocean Floor, Ridge, Shelf, Slope, Guyot



The ocean floor profile is the skeleton of every later card.

- The profile runs **land → shelf → slope → rise or trench → abyssal plain**.
- The continental **margin** is the shelf plus the slope plus the rise.
- Beyond the margin lie the abyssal plain and hills, seamounts, trenches, or the mid-ocean ridge.

Step	Depth / slope	What it is	Key fact
Continental shelf	Shore to **shelf break ~100–200 m**; slope ≤1°; **~7.5%** of ocean area; avg width ~**80 km**	Drowned edge of the continent	Rich in **fisheries and oil/gas**. World's widest shelves are **Siberian / Arctic**. India's **west shelf is wider than the east** (Gujarat–Mumbai / Khambhat up to ~300 km; Konkan–Kerala narrower). The east shelf is narrower overall but carries big **delta / Bengal Fan** sediment. Oil pair: **Bombay High** (west) and KG basin (east)
Shelf break	~130–200 m	Sudden steepening	Marks the end of the "neritic" zone
Continental slope	~200 m down to **~3000 m**; 2–5°	True edge of the continental block	Cut by **submarine canyons**. **Turbidity currents** race down them
Deep-sea fan / cone	Foot of a canyon	Turbidite dump	Same idea as an alluvial (जलोढ़) fan, underwater
Continental rise	~3000–4000 m; gentler than slope	**Sediment apron** from shelf mud and canyon fans	Present on **passive** Atlantic-type margins. Often **missing** on **active** Pacific margins where a **trench** swallows the sediment

Step	Depth / slope	What it is	Key fact
Abyssal plain	**~3000–6000 m**; almost flat	Most of the ocean floor (~**76%** of ocean area in GC framing)	Hosts red clay and Mn nodules. “Abyssal hills” are small volcanic bumps on the plain
Trench (hadal)	**>6000 m** (Mariana ~11 km)	Convergent plate where oceanic slab bends down	Island arcs or the Andes rise on the land side. Deepest points are in **N.3**
Mid-ocean ridge	Crest ~2000–3000 m (shallower than plains)	**Divergent** zone of new crust	Central **rift valley**, **transform faults**, **fracture zones**, and black smokers. Hess linked ridges to seafloor spreading

Rise vs trench - **Passive** (Atlantic type): shelf → slope → **rise** → plain - **Active** (Pacific type): shelf → slope → **trench** (little or no rise)

Seamount, guyot, atoll and related forms — Teaching Card

- A **seamount** is an undersea volcano with a **peaked / pointed** top. It rises more than about **1000 m** from the floor but does **not** break the sea surface (Emperor Seamount type).
- A **guyot** is a **flat-topped** seamount. Waves planed the summit when it was still an island; later the volcano sank and drowned. Guyots are common in the **Pacific**; coaching framing finds them in all oceans **except the Arctic**.
- Memory pair: **seamount = peaked**; **guyot = flat top** after wave bevelling and drowning.
- A **volcanic island** is a seamount that **does** break the surface (Hawaii type).
- An **atoll** is a coral reef built on a sinking volcanic peak that encloses a central **lagoon**. The classic **shape** is a **horseshoe** or **ring** around that lagoon — not a triangle or rectangle. India’s reefs need clear, warm, shallow water.
- A **submarine canyon** is a gorge cut across the **shelf and slope** — a highway for turbidity currents (Hudson Canyon type).
- A **fracture zone** is the scar of a transform fault that offsets the mid-ocean ridge.
- A **bank** is a flat, shallow elevation on a margin — **Dogger Bank** and **Grand Banks** are fish and navigation classics.
- A **hydrothermal vent** (“black smoker”) vents hot mineral fluid on volcanically active ridges.
- The **Telegraphic Plateau** is a broad high on the **North Atlantic Ridge** — not an Indian Ocean or South Atlantic land plateau.

Form	Quick recall
Seamount	Peaked undersea volcano (>1000 m)
Guyot	Flat-topped drowned seamount
Volcanic island	Seamount that breaks the surface
Atoll	Ring / horseshoe reef around a lagoon

Form	Quick recall
Submarine canyon	Gorge across shelf–slope
Bank	Flat shallow elevation (Dogger, Grand Banks)

Named ridges

Ridge	Fact
Mid-Atlantic	S-shaped; **Telegraphic Plateau** on the **North** arm
East Pacific Rise	Fast spreading
Carlsberg	NW **Indian** Ocean (Arabian Sea)
Mid-Indian / SW + SE Indian	Indian Ocean spreading
Ninetyeast	N–S volcanic trail in E Indian Ocean — **not** a classic spreading MAR

Inline PYQ — UPPCS Prelims 2019, Q77

Telegraphic Plateau is a part of the:

A. North Atlantic Ridge

B. South Atlantic Ridge

C. Indian Ocean Ridge

D. Pacific Ridge only

Show answer

Ans: A — Broad high on the **North Atlantic Ridge**.

Vertical life/depth words (do not mix with UNCLOS miles)

Word	Where
Neritic	Over the **shelf**
Bathyal	Over the **slope**
Abyssal	Over the **plain**
Hadal	Inside the **trench**

Ocean deposits (NCERT)

Deposit	Fact
Terrigenous	Land mud and sand on the shelf and slope
Red clay	Deepest abyssal deposit from volcanic and meteoric dust, with **carbonate dissolved away**

Deposit	Fact
Globigerina ooze	Calcareous foraminifera mud at mid-depth
Pteropod ooze	Calcareous ooze, shallower than globigerina
Diatom ooze	**Siliceous** ooze of cold high-latitude seas
Radiolarian ooze	Siliceous ooze of tropical deep water
Manganese nodules	Potato-shaped ores on abyssal plains — classic in the **Pacific**; India's claim is the **Central Indian Ocean Basin**

Temperature (horizontal + vertical)

- **Horizontal:** surface water is warmest near the **equator** and cools toward the poles. Warm and cold **currents** shift the pattern along coasts.
- **Vertical:** the mixed layer sits above the **thermocline (~300–1000 m)**. Deep water is near **2°C** almost everywhere.
- A **halocline** is a sharp salinity jump with depth.
- A **pycnocline** is a sharp density jump with depth.

Density (ForumIAS / NCERT link card)

- Density rises when water is **colder**, **saltier**, or under higher **pressure**.
- Cold salty water **sinks**. Warmer fresher water stays lighter. That contrast feeds **thermohaline** deep circulation.
- **Upwelling** brings deep nutrient water up. **Downwelling** is the sinking partner.

Thermohaline conveyor

- Cold, salty water sinks in the North Atlantic and around Antarctica and feeds deep global flow.
- Five subtropical **gyres** dominate the surface.
- The **Equatorial Counter Current** runs east between the two west-flowing equatorial currents — piled water from the converging equatorial currents drains eastward.

N.3 Trenches

Trenches sit on **active** (convergent) margins. Match the deepest point to the ocean.

Ocean	Deepest Point (Trench)	Exact Max Depth	Geographical Location
Pacific	**Mariana Trench** (Challenger Deep)	**11,033 m** (~36,201 ft)	Near Guam / Mariana Islands (W. Pacific)
Atlantic	**Puerto Rico Trench** (Milwaukee Deep)	**8,605 m** (up to ~9,460 m in classical texts)	North of Puerto Rico (Caribbean margin)
Indian	**Java / Sunda Trench**	**7,450–7,542 m**	South of Java & Sumatra (NE Indian Ocean)

Ocean	Deepest Point (Trench)	Exact Max Depth	Geographical Location
Arctic	**Molloy Deep** (Fram Strait)	**~5,550 m**	Between Greenland and Svalbard

- **Challenger Deep** in the Mariana Trench is about **11 km**.
- Other trench names: Tonga, **Kermadec**, Philippine, Kuril–Kamchatka / Aleutian, Peru–Chile, South Sandwich (S Atlantic), **Diamantina** (Indian Ocean fracture zone SW of Australia).

Inline PYQ — UPPCS Prelims 2022, Q114

Match List-I with List-II and select the correct answer using the code given below the lists:

List-I (Ocean)	List-II (Deepest point)
A. Pacific	1. Sunda Trench
B. Arctic	2. Puerto Rico Trench
C. Indian	3. Mariana Trench
D. Atlantic	4. Molloy Deep

A. A-1, B-2, C-3, D-4

B. A-3, B-4, C-1, D-2

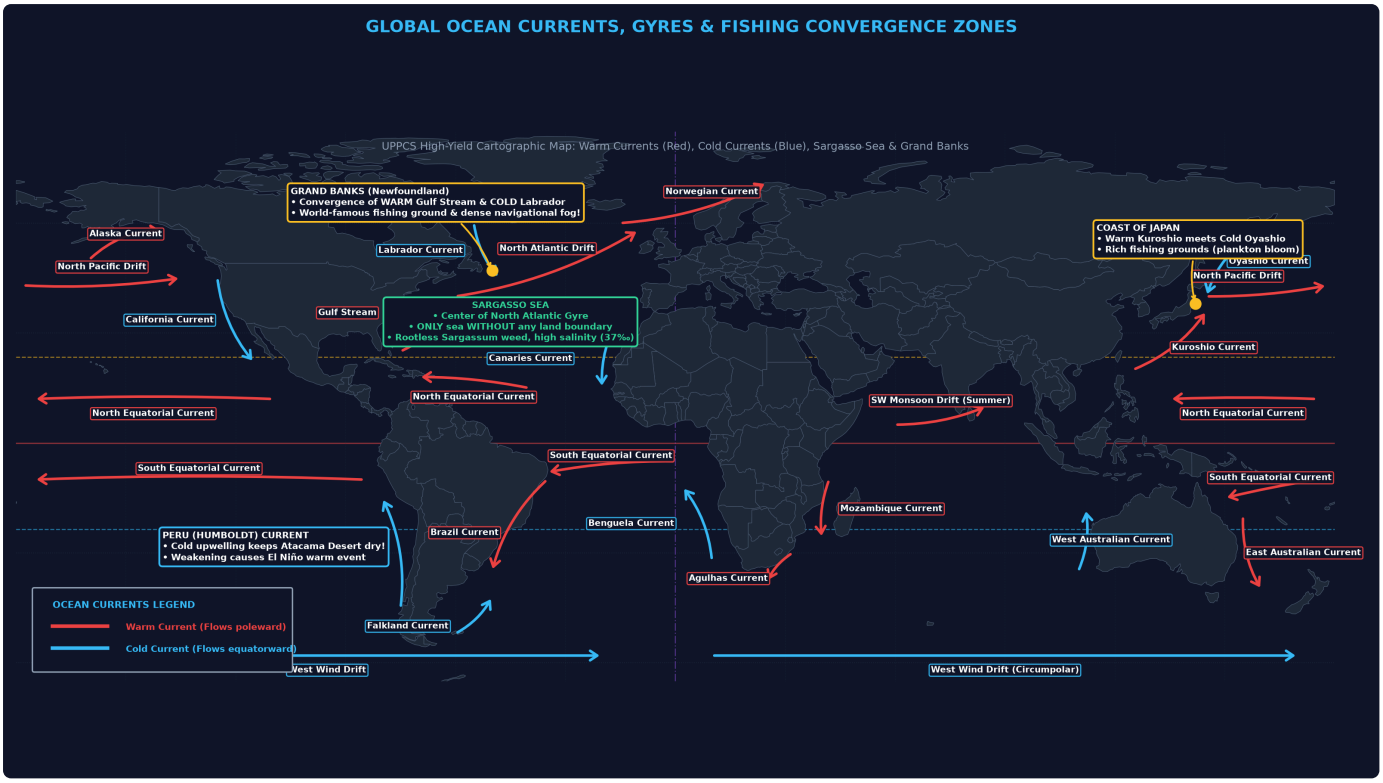
C. A-3, B-1, C-4, D-2

D. A-2, B-4, C-1, D-3

Show answer

Ans: B — Pacific–Mariana (3), Arctic–Molloy (4), Indian–Sunda (1), Atlantic–Puerto Rico (2).

N.4 Ocean Currents (warm & cold)



Ocean currents are driven by prevailing winds, the **Coriolis** force (deflection right in the Northern Hemisphere, left in the Southern) and the shape of the land.

- **Warm** currents commonly wash the **east coasts** of continents (west sides of ocean basins).
- **Cold** currents and upwelling favour the **west coasts** of continents, plus polar water. **Pacific**

Current	Type
Kuroshio (Japan)	**Warm**
North Pacific	Warm drift
Alaska	Warm
California	**Cold**
Humboldt / Peru	**Cold** — upwelling, rich fish. El Niño shuts this belt
East Australian	Warm
Oyashio / **Kurile**	**Cold**
Equatorial (N/S) + Equatorial Counter	

Atlantic

Current	Type
Gulf Stream / Florida	**Warm**

Current	Type
North Atlantic Drift	Warm drift that keeps NW Europe mild — the **Blanket of Europe**, helping keep Norway harbours milder and freer of ice
Canary	**Cold**
Labrador	**Cold** (meets Gulf Stream → Grand Banks fog/fish)
Brazil	**Warm**
Benguela	**Cold** — **SE Atlantic**, SW Africa — **not** Pacific
Falkland / Malvinas	Cold
Guinea	Warm

Indian

Current	Type
Agulhas	**Warm** current off SE Africa — the classic **Indian Ocean** option (not Florida, Canary or Kurile)
Mozambique	Warm
Somali	Seasonal (SW monsoon (दक्षिण-पश्चिम) upwelling off Somalia)
West Australian	**Cold**
Monsoon current	**Reverses** with monsoon — why N Indian Ocean current direction changes seasonally

Southern - The West Wind Drift / **Antarctic Circumpolar Current** is a **cold** current. - It is the only current that **circles the globe**.

East-coast rule: warm currents (Brazil, Agulhas, Kuroshio, Gulf Stream, East Australian).

West-coast rule: cold upwelling (Humboldt, California, Canary, Benguela).

Upwelling (cold + fish)

- Classic belts: Humboldt / Peru, California, Canary, **Benguela**, and Somalia under the SW monsoon.
- These sit on the **west coasts** of continents (east sides of ocean basins).

El Niño / La Niña (ला नीना) / IOD

- **El Niño** brings warm water **off Peru**. Upwelling stops, so **plankton and fish fall**.
- **La Niña** is the opposite — extra-cold Peru and stronger upwelling.
- The **Indian Ocean Dipole (IOD)** is a separate Indian Ocean temperature swing — not the same as El Niño.

Inline PYQ — UPPCS Prelims 2025, Q75

Which of the following are NOT cold ocean currents?

1. Agulhas Current

2. Brazil Current
3. Humboldt Current
4. California Current

A. 1 and 3

B. 3 and 4

C. 2 and 3

D. 1 and 2

Show answer

Ans: D — Agulhas and Brazil are **warm**. Humboldt and California **are** cold. Trap B is the cold pair if you miss **NOT**.

Inline PYQ — UPPCS Prelims 2020, Q78

Which of the following ocean currents is associated with the Indian Ocean?

A. Florida Current

B. Canary Current

C. Agulhas Current

D. Kurile Current

Show answer

Ans: C — **Agulhas** (warm, SE Africa).

Inline PYQ — UPPCS Prelims 2023, Q67

With reference to El Niño, which of the following statements is/are correct?

1. A warm ocean current appears off the Peru coast.
2. Due to El Niño, the amount of plankton increases.

A. Only 1

B. Only 2

C. Both 1 and 2

D. Neither 1 nor 2

Show answer

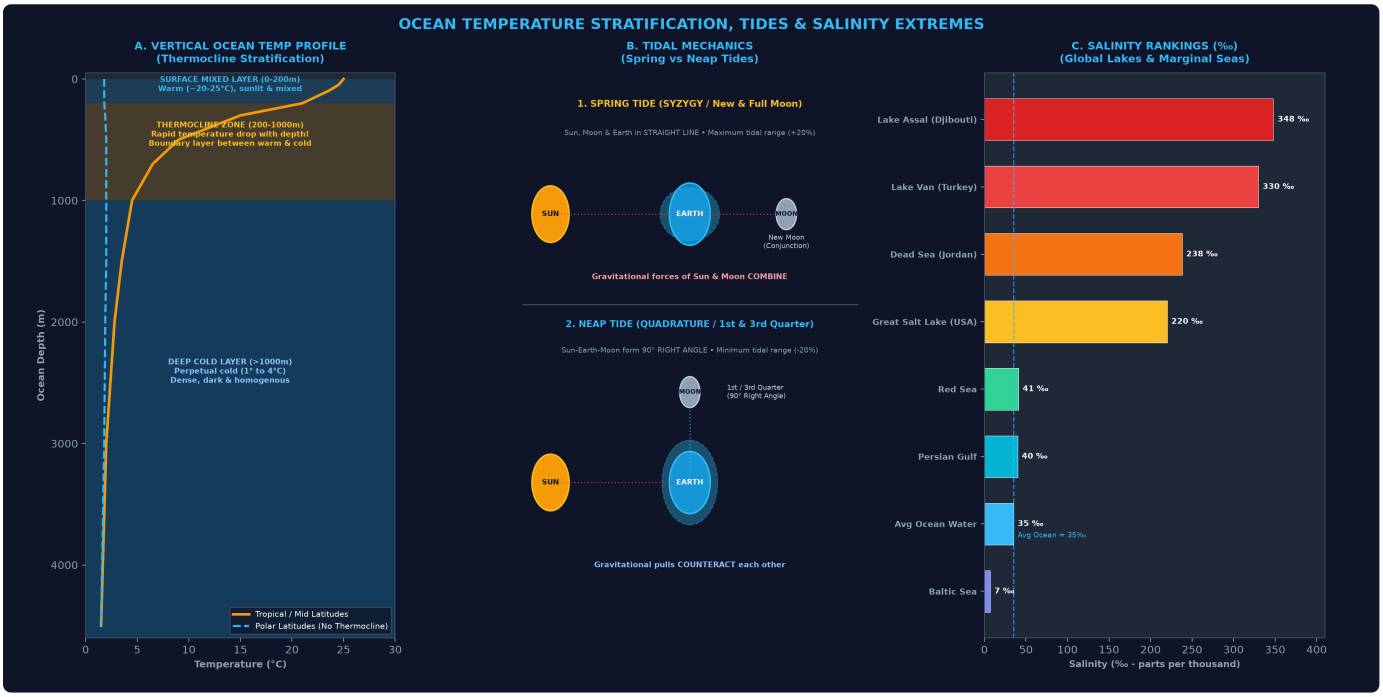
Ans: A — Warm water **cuts upwelling**, so plankton **fall**.

Extra Atlantic names: the **Irminger** and Norwegian currents are warm. The East Greenland Current is cold.

Effects of ocean currents

- **Climate:** warm currents raise coastal temperatures. The Gulf Stream and North Atlantic Drift keep NW Europe mild. Cold currents cool coasts and can help desert belts along west coasts (Atacama, Namib, Peru–Chile, Canary–Sahara fringe).
- **Fog and fisheries:** where **warm and cold currents meet**, fog and rich fishing grounds appear. The Grand Banks mix the Gulf Stream with the Labrador Current. Seas near Japan mix the Kuroshio with the Oyashio.
- **Upwelling coasts** (Peru, California, Benguela, Canary, Somalia) are plankton- and fish-rich until El Niño shuts Peru upwelling.
- **Navigation:** currents speed or slow ships. Knowledge of gyres mattered historically and still matters for routing.

N.5 Salinity



Open-ocean salinity averages about **35‰** (35 g salt per 1000 g seawater). **Sodium chloride (NaCl)** is about **77%** of the dissolved salts in the dissolved-salt table. Controls are evaporation, rainfall, river inflow and ice freeze / melt.

Belt	Salinity
Tropics of Cancer / Capricorn	**Maximum** in the open ocean — high evaporation, less rain
Equator	Lower than the tropics (heavy rain)
Poles	Lower (ice melt, low evaporation)

- **High** seas: Red Sea, Persian Gulf, Mediterranean.
- **Low** seas: Baltic, equatorial belt, big river mouths.
- Dead Sea and Great Salt Lake are **lakes**, not ocean MCQs.

Hypersaline lake	Approx. salinity	Place note
Lake Assal	**~348‰**	Often cited as among the world's highest lake salinities
Lake Van	**~330‰**	**Türkiye**
Dead Sea	**~238‰**	Jordan Rift; classic high-salinity lake
Great Salt Lake	**~220‰**	**Utah, USA**

- A **halocline** is a depth zone where salinity changes rapidly.
- A **thermocline** is a depth zone where temperature changes rapidly.
- A **pycnocline** is a depth zone where density changes rapidly.
- Open-ocean NaCl ≈ **77.8%** of dissolved salts (Dittmar / Challenger chemistry note).
- Arabian Sea salinity often higher than Bay of Bengal (~36‰ vs ~30‰) — more evaporation, fewer big rivers.

Inline PYQ — UPPCS Prelims 2023, Q61

Which of the following statements is true?

A. Benguela Current is a cold current of the Pacific Ocean.

B. If the Sun, the Earth and the Moon are in a straight line, then this situation results in a small tide.

C. Ocean salinity is maximum near the Tropics of Cancer and Capricorn.

D. Tide comes on the Earth every day after exactly 12 hours 30 minutes.

Show answer

Ans: C — A is false (Benguela = **Atlantic**). B is false (syzygy = **spring / large**). D is false (period ≈ **12 h 25 min**).

N.6 Tides

The Moon's pull exceeds the Sun's. A lunar day is about **24 h 50 min**, so successive high tides are about **12 h 25 min** apart — **not** exactly 12 h 30 min.

Tide types — Teaching Card

Spring vs neap (वृहत्) (geometry)

- A **spring tide** occurs when the Sun, Earth and Moon lie roughly in a **straight line** (new moon or full moon).
- Lunar and solar pulls then **add**, so the high tide is **extra high** and the low tide is **extra low** — the tidal range is **large**.
- A **neap tide** occurs at **quadrature**, when the Moon is at first or third quarter and the Sun–Earth–Moon angle is about **90°**.
- Lunar and solar pulls then **partly cancel**, so the tidal range is **small**.

- Spring tides are **not** named for the season — they mean a strong (“springing”) range.

Daily rhythm

- A **semi-diurnal** tide has **two** high waters and **two** low waters in about one lunar day — the most common world pattern.
- A **diurnal** tide has **one** high and **one** low per lunar day.
- A **mixed** tide shows two highs and two lows of **unequal** height.

Type	Quick recall
Spring	Syzygy (new/full); **large** range
Neap	Quadrature (quarters); **small** range
Semi-diurnal	2 high + 2 low
Diurnal	1 high + 1 low
Mixed	Unequal highs / lows

- The highest tidal range in the world is the **Bay of Fundy** (Canada).
- India’s tidal-energy coast fact is the **Gulf of Khambhat** (ahead of Kutch in standard options).

Tidal bore

- A bore is a wall of tide in a funnel estuary (ज्वारनदमुख).
- India’s classic bore is the **Hooghly**. World pairs include the Amazon and China’s Qiantang.
- **Flood tide** is the rising tide. **Ebb tide** is the falling tide.
- A perigean spring is an extra-large spring tide when the Moon is at perigee.
- At **neap**, high tide sits **below** normal spring height and low tide sits **above** normal low. Neaps are **not** at new or full moon (those are springs). Neaps fall near day 7–8 of the lunar fortnight.
- The indirect high tide on the far side of Earth is linked to **centrifugal** balance with Moon gravity.
- **Southampton** (England) is famous for **four** (चातुर्याम) tides a day because of Channel and North Sea timing.
- The Moon’s tide-raising force is about **2.17×** the Sun’s.

Waves (ForumIAS: teach separately from tides)

- Ordinary waves need wind and **fetch**. Energy travels. Water itself mostly oscillates.
- **Constructive** waves have a strong swash and help **build** a beach.
- **Destructive** waves have a strong backwash and **erode** the beach or cliff.
- A **tsunami** is a seismic sea wave — **not** a tide and not an ordinary wind wave.

Inline PYQ — UPPCS Prelims 2022, Q20

Which of the following coasts of India is most suitable for tidal energy generation in standard options?

A. Gulf of Kutch only always

B. Gulf of Khambhat

C. Coromandel only

D. Konkan only without gulf funnel

Show answer

Ans: B — **Gulf of Khambhat**.

N.7 Straits

A strait is a narrow natural waterway joining two larger water bodies. Always ask **which two seas** it joins, and **which countries** hold the shores.

Strait of Hormuz — must-score chokepoint

Persian Gulf ↔ Gulf of Oman → Arabian Sea.

Item	Fact
Shores	**Iran** on the north; **Oman** (Musandam Peninsula) on the south
Nearby	**UAE** maritime zones sit on the Gulf and Arabian Sea **approaches**, but the classic shore pair is Iran–Oman
Width	About **21–29 nm** at the narrowest teaching band; shipping lanes are only a few miles wide each
Oil	Roughly **20 mb/d** / about **25%** of world **seaborne oil** in recent IEA figures
Gas	About **20%** of world **LNG** (mainly **Qatar** and **UAE**)
Gulf states behind the gate	Iran, Iraq, Kuwait, Saudi Arabia, Bahrain, Qatar, UAE — **Oman is outside** the Persian Gulf coast list
Bypass	Only Saudi Arabia and the UAE have useful pipelines that can skip Hormuz; the others are almost fully dependent on the strait
Buyers	Most cargo goes to **Asia** (China, India, Japan, Korea)

2026 news frame: From late February 2026, conflict around Iran turned Hormuz into a live chokepoint story — restricted transit, tanker risk, insurance spikes, and talk of dual blockades. Geography stays the same: **only sea door** out of the Persian Gulf.

Other high-yield straits — seas joined + shores

Strait	Joins	Shore / country cue
Hormuz	Persian Gulf ↔ Gulf of Oman	**Iran–Oman**
Bab-el-Mandeb	Red Sea ↔ Gulf of Aden	**Djibouti / Eritrea** side ↔ **Yemen**; gate to the Indian Ocean for Suez traffic

Strait	Joins	Shore / country cue
Malacca	Andaman / South China Sea approaches ↔ Indian Ocean	**Malaysia–Indonesia** (Singapore at the tip)
Gibraltar	Mediterranean ↔ Atlantic	**Spain–Morocco** (UK: Gibraltar Rock)
Bosphorus	Black Sea ↔ Marmara	Entirely **Türkiye**
Dardanelles	Marmara ↔ Aegean	Entirely **Türkiye**
Bering	Arctic ↔ Pacific	**Russia–USA** (Alaska)
Palk	Bay of Bengal side ↔ Gulf of Mannar / Laccadive approaches	**India–Sri Lanka**
10° Channel	Andaman ↔ Nicobar (निकोबार)	Inside India
9° Channel (नौ डिग्री)	Minicoy ↔ Lakshadweep (लक्षद्वीप) proper	Inside India
8° Channel	Maldives ↔ Minicoy	Maldives–India
Duncan Passage	Little Andaman ↔ South Andaman	Inside India
Sunda	Indian Ocean ↔ Java Sea approaches	**Sumatra–Java** (Indonesia)
Torres	Coral / Arafura approaches	**Australia–Papua New Guinea**
Bass	Tasman Sea approaches	**Australia–Tasmania**
Magellan	Atlantic ↔ Pacific via Chilean channels	**Chile**
Drake Passage	Atlantic ↔ Pacific (open water)	South of Cape Horn — **not** Magellan
Dover	North Sea ↔ English Channel	**UK–France**
Mozambique	Indian Ocean corridor	**Mozambique–Madagascar**
Davis	Baffin Bay approaches	**Greenland–Canada**
Denmark Strait	Greenland Sea / Atlantic	**Greenland–Iceland**
Cook	Pacific between NZ islands	North Island ↔ South Island
Florida / Yucatan	Gulf of Mexico gates	USA–Cuba / Mexico–Cuba approaches
Formosa / Taiwan	East China Sea ↔ South China Sea	Taiwan Strait
Korea / Tsushima	Japan Sea approaches	**Japan–Korea**
Lombok / Makassar	Indonesian passages	Tanker alternatives when Malacca is tight
Kerch	Black Sea ↔ Sea of Azov	**Russia–Ukraine** zone
Otranto	Adriatic ↔ Ionian	Italy–Albania approaches

Strait	Joins	Shore / country cue
Skagerrak / Kattegat	North Sea ↔ Baltic approaches	Scandinavia–Denmark gate

Always ask which two larger seas the strait joins.

N.8 Canals & Waterways

A **canal** is a dug waterway. A **strait** is a natural narrows. For every canal, fix three things: **which seas it joins**, **opened year**, and **whether ships must climb**.

The one contrast that clears most traps

	Suez	Panama
Opened	**1869**	**1914**
Country	**Egypt**	**Panama**
Joins	**Mediterranean ↔ Red Sea**	**Atlantic / Caribbean ↔ Pacific**
Ship height	**Sea-level** — water stays roughly one height	Ships **climb stepped chambers** over the isthmus
Famous lake	Canal lakes (Manzala…Bitter)	**Gatun Lake** (fresh reservoir on the climb)
Shortcut avoids	**Cape of Good Hope** (Africa)	**Cape Horn** (South America)
India–Europe	Cut about **7000 km**	Not the India–Europe Med route

Suez Canal — teach the full route

Mediterranean (north) → Red Sea (south).

- North mouth town: **Port Said**.
- Middle town: **Ismailia**.
- South mouth town: **Suez** (on the Gulf of Suez / Red Sea).
- Length teaching band: about **160–190 km**.
- The canal itself does **not** touch the **Atlantic** or the open **Indian Ocean**.
- From Europe you enter at Port Said; from Asia / India you enter via the Red Sea at Suez.
- After Suez, the ship still sails the Red Sea → Bab-el-Mandeb → Gulf of Aden → Arabian Sea / Indian Ocean. So “Suez connects Indian Ocean” is a **route story**, not the canal’s two end seas.
- UPPCS 2025 water-body pair = **Red Sea + Mediterranean** only.

Lakes north → south (must order)

Say: **“Man Takes Great Little.”**

1. **Manzala** — near Port Said / Mediterranean
2. **Timsah** — near Ismailia
3. **Great Bitter Lake**
4. **Little Bitter Lake** — toward Suez / Red Sea

Do **not** mix these with the natural **Gulf of Suez** (NW arm of the Red Sea). The gulf is the Red Sea inlet; the lakes sit **inside** the dug canal.

Why it mattered for India

Before 1869, Europe–India steamers often rounded **Africa**. Suez cut that maritime distance by about **7000 km**.

Panama Canal — climb, don't confuse with Suez

Atlantic / Caribbean (north-east side) → Pacific (south-west side).

- Atlantic-side town cue: **Colón**.
- Pacific-side town cue: **Panama City**.
- Ships enter gated **stepped chambers**, rise to **Gatun Lake**, then descend to the other ocean.
- It is **not** Med ↔ Red. It does **not** shorten the classic India–Europe Suez route.
- Its job is East Coast Americas / Atlantic trade (पण्याध्यक्ष) ↔ Pacific without rounding **Cape Horn**.

Other canals (match-list desk)

Canal	Joins	Country / note
Kiel	**North Sea** ↔ **Baltic**	**Germany** (Elbe estuary side ↔ Baltic). Not Mediterranean
Corinth	Ionian ↔ Aegean	Greece
Soo (Sault Ste. Marie)	Lake **Superior** ↔ **Huron**	USA–Canada. **Not** Superior–Michigan
Welland	Erie ↔ Ontario	Bypasses Niagara Falls
Erie Canal	Lake Erie ↔ Hudson River	Historic (ऐतिहासिक) US inland route
St Lawrence (लॉरेंस) Seaway	Great Lakes ↔ Atlantic	USA–Canada ocean door
Rhine–Main–Danube	North Sea inland system ↔ **Black Sea**	Europe cross-continent inland
White Sea–Baltic	Baltic ↔ White / Arctic approaches	Russia
Ludwig's	S Germany inland waterway	Match-list filler
Volga–Don	Caspian approaches ↔ Black Sea / Azov system	Russia

India inland

- **NW-1** is the Ganga–Bhagirathi (भागीरथी)–Hooghly route from **Haldia to Prayagraj**.

Inline PYQ — UPPCS Prelims 2025, Q138

The Suez Canal connects which of the following?

1. Atlantic Ocean
2. Red Sea
3. Mediterranean Sea
4. Indian Ocean

A. 1 and 3

B. 3 and 4

C. 2 and 3

D. 1 and 2

Show answer

Logic: Stem lists four water bodies. Suez's own ends are only **Mediterranean** and **Red Sea**. Atlantic is wrong. Indian Ocean is reached **after** the Red Sea, so it is not a canal-end pair in this paper. **Ans: C.** Ends = **2 Red Sea** and **3 Mediterranean**. Trap: picking Indian Ocean because India–Europe ships use Suez as a **route**; the canal itself stops at the Red Sea.

Inline PYQ — UPPCS Prelims 2019, Q76

Which of the following is the correct sequence of lakes of the Suez Canal from north to south?

A. Timsah — Manzala — Great Bitter — Little Bitter

B. Great Bitter — Little Bitter — Timsah — Manzala

C. Little Bitter — Great Bitter — Timsah — Manzala

D. Manzala — Timsah — Great Bitter — Little Bitter

Show answer

Logic: North→south lake order starts at the Mediterranean end. Mnemonic **"Man Takes Great Little"** = Manzala → Timsah → Great Bitter → Little Bitter. **Ans: D.** **Manzala → Timsah → Great Bitter → Little Bitter**. Trap: starting with Timsah or reversing from the Red Sea end.

Inline PYQ — UPPCS Prelims 2018, Q29

After the opening of the Suez Canal, the distance between India and Europe was reduced by about:

A. 5,000 km

B. 7,000 km

C. 8,000 km

D. 10,000 km

Show answer

Logic: Classic distance cut after Suez opened. Coaching / UPPCS figure is about **7000 km**, not 5000 / 8000 / 10000. **Ans: B.** About **7000 km**. Trap: neighbouring round numbers in the options.

N.8A Coral reefs

Coral reefs are tropical shallow-water communities built by coral polyps and their symbiotic algae. Coaching often calls them the “rainforests of the ocean” because they host huge biodiversity on a tiny share of ocean floor.

Conditions

- Reefs grow mainly between the **Tropics** (roughly **30°N–30°S**).
- Mean water temperature is usually about **20–30°C**.
- Salinity must stay moderate (नरम दल). Fresh floodwater and extreme salinity both injure polyps.
- Depth is shallow enough for sunlight — commonly within about **45–55 m** — so zooxanthellae (ज़ूज़ैंथेली) can photosynthesize.
- Water must be **clear and sediment-free**. Muddy river mouths are poor reef sites.

Reefs cover only a tiny share of ocean floor but host a huge share of marine species (often cited near **~25%**).

Coral bleaching (प्रवाल विरंजन) is the loss of zooxanthellae under heat stress. The coral turns white. If stress lasts too long, the colony (उपनिवेश) can die. Warming seas are the main modern trigger; pollution and sedimentation add stress.

Great Barrier Reef is the world’s largest reef system. It lies off **NE Australia / Queensland** in the **Pacific** and stretches about **2300 km**. Coaching figures often list roughly 400 coral types, 1500 fish types and 4000 mollusc types.

Reef types include **fringing** reefs close to shore, **barrier** reefs offshore with a lagoon, and **atolls** as **ring or horseshoe** reefs around a lagoon. The **shape of an atoll** is therefore **horseshoe / ring**, not triangular or rectangular. **Addu Atoll** in the southern Maldives sits in the **Indian Ocean**. India’s own reef tracts are taught with the biodiversity chapter (Andaman–Nicobar (अंडमान), Lakshadweep, Gulf of Mannar, Gulf of Kachchh).

N.9 UNCLOS, Territorial Waters, EEZ, Shelf

UNCLOS was adopted (अंगीकृत) in **1982**, entered into force in **1994**, and India ratified in **1995**. Zones are measured from **baselines**.

Zone	Breadth
Territorial sea	**12 nautical miles** — full sovereignty, with innocent passage
Contiguous zone	**24 nm** — customs, fiscal, immigration and sanitary control
EEZ	**200 nm** — sovereign rights to resources (fish, oil, wind)
Continental shelf	At least **200 nm**, and may extend to **350 nm** if geology allows
High seas	Beyond national zones
The Area	Seabed beyond national jurisdiction — **common heritage**, managed by the **ISA** (Kingston, Jamaica)

Passage

- **Innocent passage** applies in the territorial sea.
- **Transit passage** is the stronger right through international straits.
- Archipelagic waters are the Indonesia / Philippines type.

India

- EEZ is about **2.02 million km²**.
- Coastline with islands is **7,516.6 km**.
- Nodule claim: **Central Indian Ocean Basin**.

BBNJ 2023

- High-seas biodiversity treaty; India signed.
- It does **not** change the 12 / 24 / 200 nm yardsticks.

N.10 Important Fisheries

Major **fisheries** thrive where the continental **shelf is wide**, where cold and warm currents mix, or where **upwelling** lifts nutrient-rich deep water into the sunlit layer. Nutrients feed plankton. Plankton feed fish. Narrow, steep shelves and nutrient-poor warm pools are weaker fishing grounds.

Ground	Why
Grand Banks (Newfoundland)	Labrador cold + Gulf Stream
Dogger Bank	North Sea shelf
Peru–Chile	**Humboldt upwelling** (fails in El Niño)
Japan / Kurile	Kuroshio + Oyashio mix
Iceland / Norway	Arctic/Atlantic mix
India	West coast and Wadge Bank, under EEZ rights

The **Grand Banks** off Newfoundland are rich because the cold **Labrador Current** meets the warm **Gulf Stream**. **Dogger Bank** is a shallow North Sea shelf ground. The **Peru–Chile** coast depends on **Humboldt upwelling**; that upwelling weakens in **El Niño**, so the fishery collapses. Seas near **Japan** and the **Kuriles** mix the warm **Kuroshio** with the cold **Oyashio**. Waters around **Iceland** and **Norway** mix Arctic and Atlantic water on a productive shelf. India's best grounds sit on the wider **west coast** shelf and on **Wadge Bank** south of India, within the national **EEZ**.

N.11 Polar Regions & Antarctic Treaty

The Arctic is an ocean with sea ice and rim lands. Antarctica is a **continent** under an ice sheet.

	Arctic	Antarctic
Nature	Ocean + sea ice + rim lands	**Continent** + ice sheet
India station	**Himadri** (Ny-Ålesund, Svalbard)	**Maitri**, **Bharati**

Antarctic Treaty

- Signed **1 Dec 1959** in Washington. It entered into force in **1961**.
- The original **12** parties agreed on peaceful use, science, no new claims, and no military bases for warfare.
- India acceded in **1983** (consultative status).
- The **Madrid Protocol 1991** adds environment (पर्यावरण) rules and a **mining ban**.

Stations

- **Dakshin Gangotri** (1983/84) was India's first base and is **not** the operational pair today.
- Operational Antarctic stations are **Maitri** (1988/89, Schirmacher) and **Bharati** (2012, Larsemann Hills).

Arctic

- **Himadri** sits at Ny-Ålesund, Svalbard (2008).
- India is an **observer (2013)** in the Arctic Council, not a full member.
- India's Arctic Policy dates to **2022**. **IndARC** is a moored observatory.

Inline PYQ — UPPCS Prelims 2021, Q53

Which of the following are India's permanent research stations in Antarctica?

A. Dakshin Gangotri and Himadri

B. Bharati and Maitri

C. Himadri and Maitri

D. Dakshin Gangotri and Bharati

Show answer

Ans: B — Operational = **Bharati and Maitri**. Himadri is **Arctic**.
